

■ ■ ISRO SPACE APPLICATIONS CENTRE (SAC) — AHMEDABAD

SPACEFLIGHT MICROELECTRONICS QUALIFICATION CERTIFICATE  
adhering to MIL-STD-883 Method 1015 | MIL-PRF-38535 Class V | ESA ECSS-Q-ST-60C

1. Component Identity & Environmental Test Parameters

|                      |                           |                    |                      |
|----------------------|---------------------------|--------------------|----------------------|
| Component Serial ID: | ISRO-SAC-2026-0001        | Test Chamber Temp: | 125.0°C (HTOL)       |
| Device Family:       | DIGITAL_IC                | Vacuum Pressure:   | 10■ Torr (TVAC)      |
| Project Flight Code: | ISRO_GAGANYAAN_SAC_LOT_01 | ATE Station ID:    | SAC-AHMEDABAD-ATE-01 |
| Screening Parameter: | Quiescent Current (µA)    | Specification USL: | 45.0 µA              |

2. Telemetry Measurements & 95% Conformal Uncertainty Interval

| Measurement Checkpoint   | Parametric Value | Robust Z-Score | Conformal 95% Bounds  |
|--------------------------|------------------|----------------|-----------------------|
| 0h Baseline Measurement  | 11.20 µA         | 0.00 σ         | N/A (Pre-Stress Base) |
| 24h Chamber Telemetry    | 12.10 µA         | 0.45 σ         | Live Telemetry Sensor |
| 168h Forecast Trajectory | 14.80 µA         | 0.81 σ         | [13.50 - 16.10] µA    |

3. Game-Theoretic SHAP Physics Feature Attribution Breakdown

SHAP (Shapley Additive exPlanations) uses cooperative game theory to quantify the exact marginal contribution of each physical stressor and parametric variable to the forecasted 168-hour degradation trajectory:

| Physical Feature / Kinetic Variable      | Observed Value | SHAP Attribution Value | Governing Physics Failure Mechanism    |
|------------------------------------------|----------------|------------------------|----------------------------------------|
| 24h Current Measurement (X_24h)          | 12.10 µA       | +0.715 µA              | Arrhenius Thermal Leakage Base         |
| Kinetic Degradation Velocity (v24)       | 0.0375 µA/hr   | +0.469 µA              | Black's Electromigration / Oxide Drift |
| Robust Population Z-Score (Z_mad)        | 0.45 σ         | +0.189 µA              | Wafer Periphery Process Gradient       |
| Arrhenius Activation Factor (Ea=0.68 eV) | 125.0°C Stress | +0.250 µA              | Shockley-Read-Hall (SRH) Trap Gen      |

4. Quality Assurance (QA) Engineer Decision Audit Report

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ■ QUALIFIED FOR 24H FLIGHT RELEASE (AUTO-PASS)                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>QA INSPECTION SUMMARY — NOMINAL DEGRADATION:<br/>Component ISRO-SAC-2026-0001 exhibits nominal Arrhenius thermal degradation kinetics (<math>v_{24} = 0.0375\ \mu\text{A/hr}</math>). The non-parametric 95% Conformal Prediction Upper Bound (16.10 µA) is safely below the Specification USL (45.0 µA) with a 64.2% safety margin. QA Recommendation: Approve immediate 24-hour early release from burn-in chamber. Safe for spaceflight integration.</div> |

5. ISRO QA Inspector Sign-Off & Cryptographic Verification

|                                   |       |                             |                     |
|-----------------------------------|-------|-----------------------------|---------------------|
| Lead QA Reliability Engineer:     | _____ | Date & Stamp:               | _____               |
| ISRO SAC Qualification Authority: | _____ | SHA-256 Audit Verification: | a7f9b2c4e18d3091... |